# 基于 STM32 的语音交互式多功能检测终端

# 部署简易手册

# 1 开发环境

## 1.1 硬件列表

| 序号 | 硬件 | 型号 | 备注 |
|---|---|---|---|
| 1 | 主控芯片 | STM32F411RET6 | |
| 2 | WiFi 模块 | ESP32-S3 | |
| 3 | 语音模块 | ASRPRO | |
| 4 | 传感器模块 | AHT21<br>SPL06-001<br>LSM303DLH<br>MPU6050<br>MAX30102<br>LSM303DLHC | |
| 5 | 显示 | TFT LCD | 驱动芯片为 ST7789V |
| 6 | 麦克风 | INMP441 | |
| 7 | 功放 | MAX98357 | |
| 8 | ST-Link | | 用于 STM32 的程序烧录和代码调试 |

## 1.2 软件环境

| 软件名称 | 版本 | 备注 |
|---|---|---|
| Keil MDK | v5 | STM32 程序开发与编译 |
| STM32CubeMX | v6.12 | 外设初始化配置 |
| FreeRTOS | v10.3.x | 系统任务调度 |
| LVGL | v8.x | 图形界面开发 |
| Vscode、ESP-idf | v1.85.x、v1.10.x | esp32 开发环境 |
| 串口调试助手 | | 串口数据调试 |
| 天文 Block | 2025 | 语音模块编程软件 |

# 2 终端硬件连接列表

| 硬件模块 | 自身引脚 | 连接对象 | 连接对象引脚 | 备注 |
|---|---|---|---|---|

| 硬件模块 | 自身引脚 | 连接对象 | 连接对象引脚 | 备注 |
| --- | --- | --- | --- | --- |
| STM32 | PA6 | IIC 模块 | SCL | IIC 时钟线 |
| STM32 | PA7 | IIC 模块 | SDA | IIC 数据线 |
| STM32 | PA3 | LED | - | LED 控制引脚 |
| STM32 | 5V | ASRPRO | 5V | 供电 |
| STM32 | GND | ASRPRO | GND | 共地 |
| STM32 | PC6 (TX) | ESP32-S3 | PG19 | 串口发送 |
| STM32 | PC7 (RX) | ASRPRO | PB5 | 串口接收 |
| ESP32-S3 | 3.3V | INMP441 | VDD | 麦克风供电 |
| ESP32-S3 | GND | INMP441 | GND | 共地 |
| ESP32-S3 | GPIO42 | INMP441 | SD | 麦克风数据引脚 |
| ESP32-S3 | GPIO1 | INMP441 | WS | 麦克风帧同步 |
| ESP32-S3 | GPIO2 | INMP441 | SCK | 麦克风时钟 |
| ESP32-S3 | V5IN | MAX98357 | Vin | 音频模块供电 |
| ESP32-S3 | GND | MAX98357 | GND | 共地 |
| ESP32-S3 | GPIO7 | MAX98357 | LRC | 音频帧同步 |
| ESP32-S3 | GPIO6 | MAX98357 | BCLK | 音频时钟 |
| ESP32-S3 | GPIO5 | MAX98357 | DIN | 音频数据输入 |

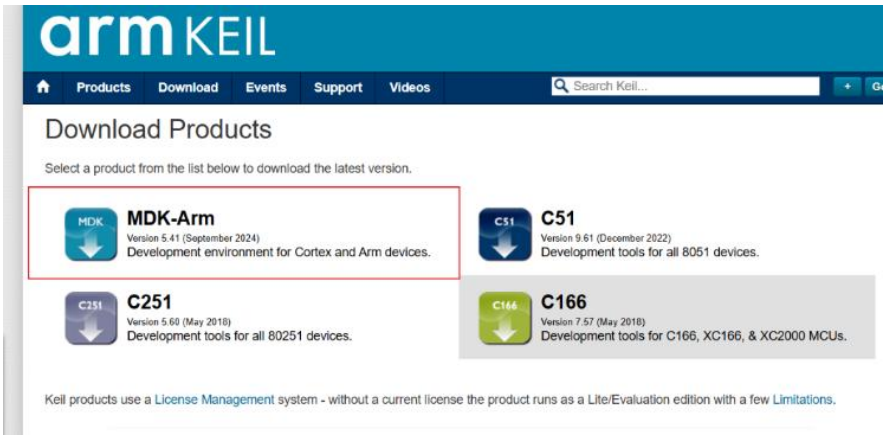
# 3 软件安装

## 3.1 MDK 安装

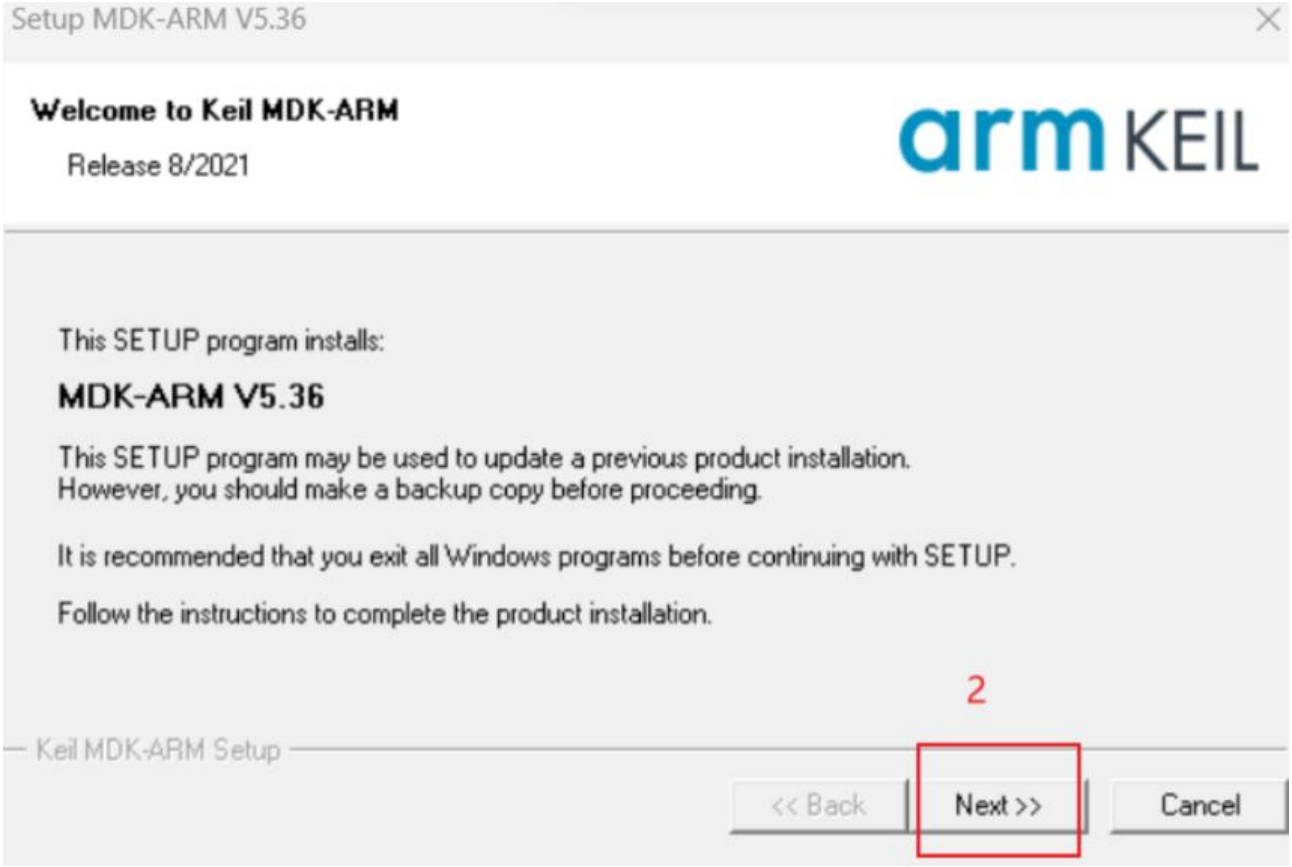
### 3.1.1 MDK 软件安装

MDK 安装 = MDK 软件安装 + 器件支持包。

MDK 软件下载地址：https://www.keil.com/download/product/



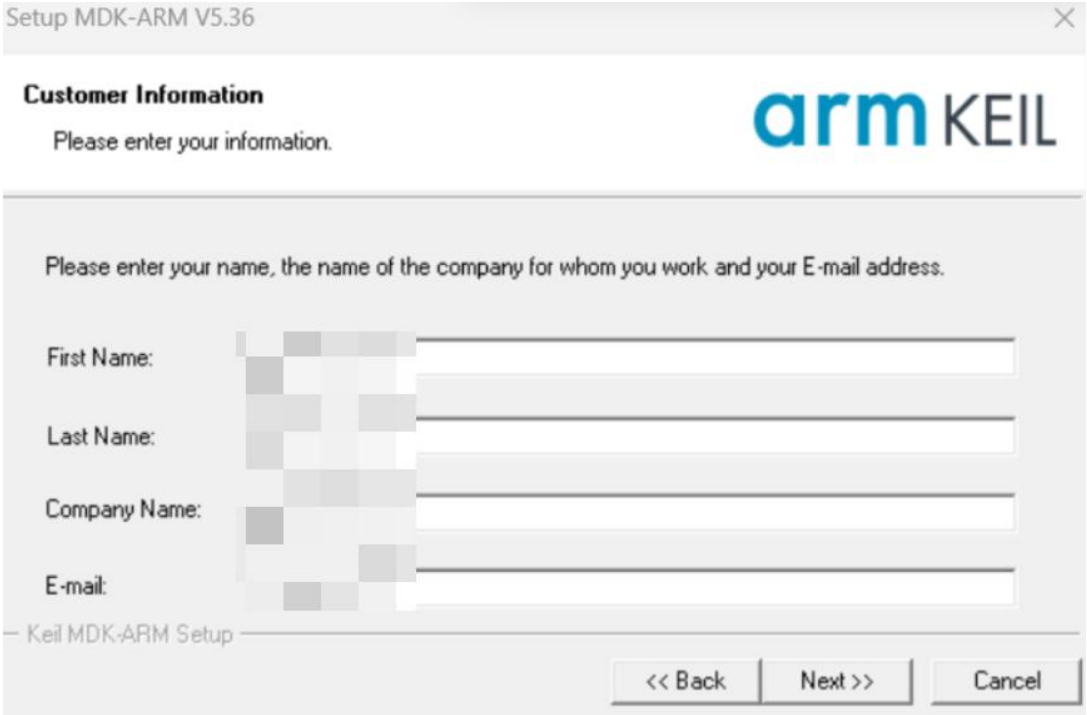
下载 mdk 的 exe 文件后，双击启动


Setup MDK-ARM V5.36
Welcome to Keil MDK-ARM
Release 8/2021
arm KEIL
This SETUP program installs:
MDK-ARM V5.36
This SETUP program may be used to update a previous product installation.
However, you should make a backup copy before proceeding.
It is recommended that you exit all Windows programs before continuing with SETUP.
Follow the instructions to complete the product installation.
2
Keil MDK-ARM Setup
<< Back
Next >>
Cancel

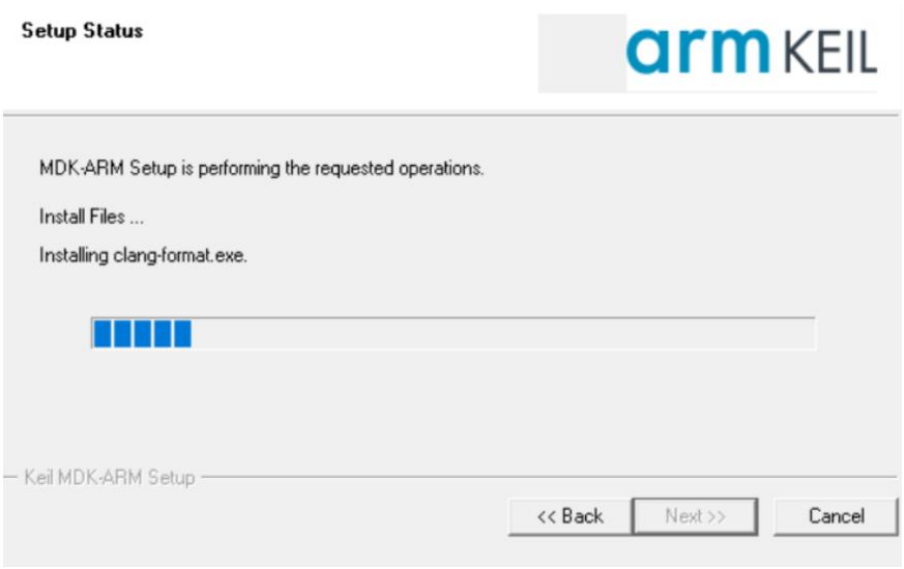
点击 next

Setup MDK-ARM V5.36
License Agreement
Please read the following license agreement carefully.
arm KEIL
To continue with SETUP, you must accept the terms of the License Agreement. To accept the agreement, click the check box below.
END USER LICENSE AGREEMENT FOR ARM SOFTWARE DEVELOPMENT TOOLS
This end user license agreement ("License") is a legal agreement between you (a single individual), or the company or organisation (a single legal entity) that you represent and have the legal authority to bind, and Arm relating to use of the Arm Tools. Arm is only willing to license the Arm Tools on condition that you accept all of the terms of this License. By clicking "I Agree" or by installing or
I agree to all the terms of the preceding License Agreement
勾选
Keil MDK-ARM Setup
<< Back
Next >>
Cancel
next

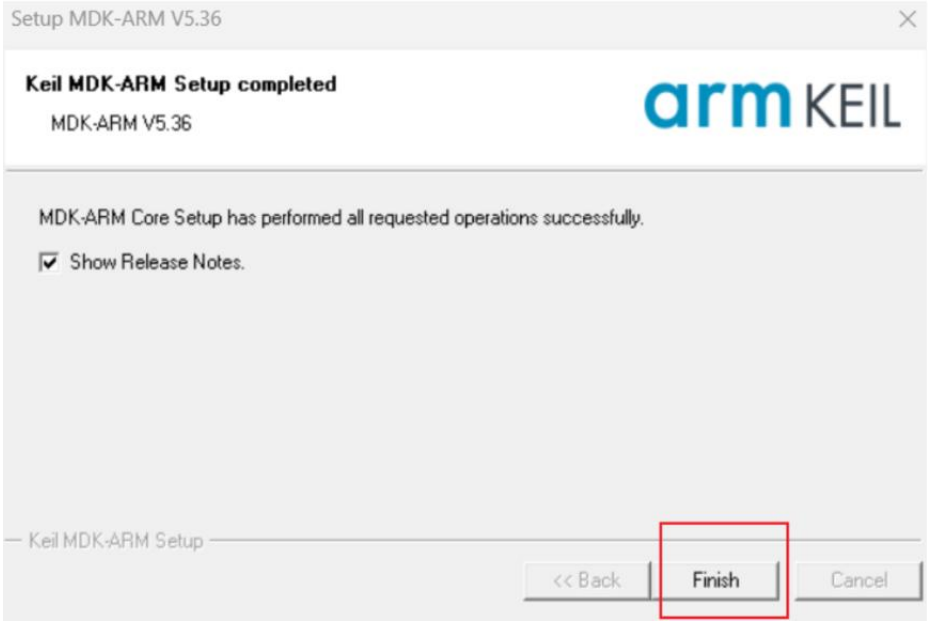
根据需要选择安装的目录




根据实际情况填写信息后，点击 next



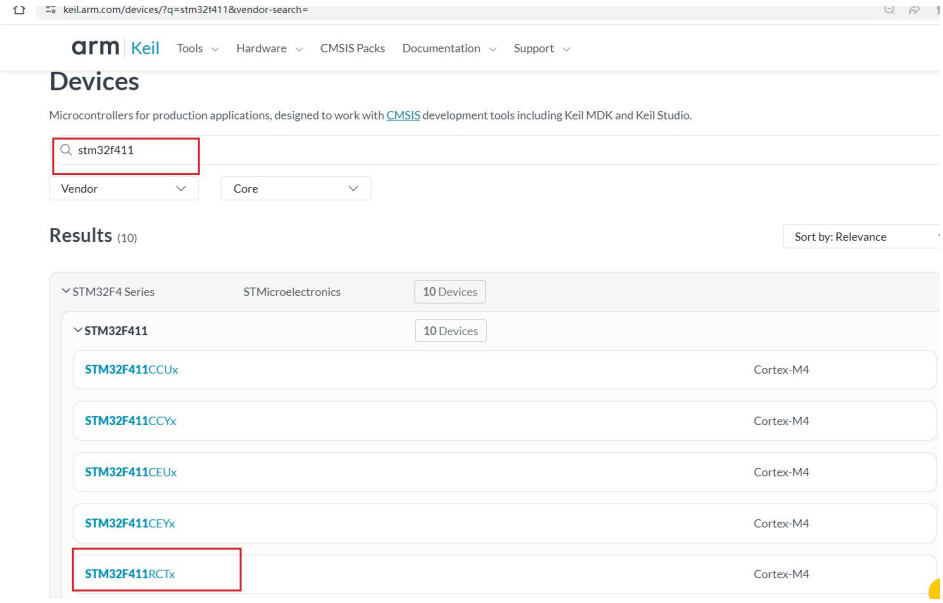

等待安装完成



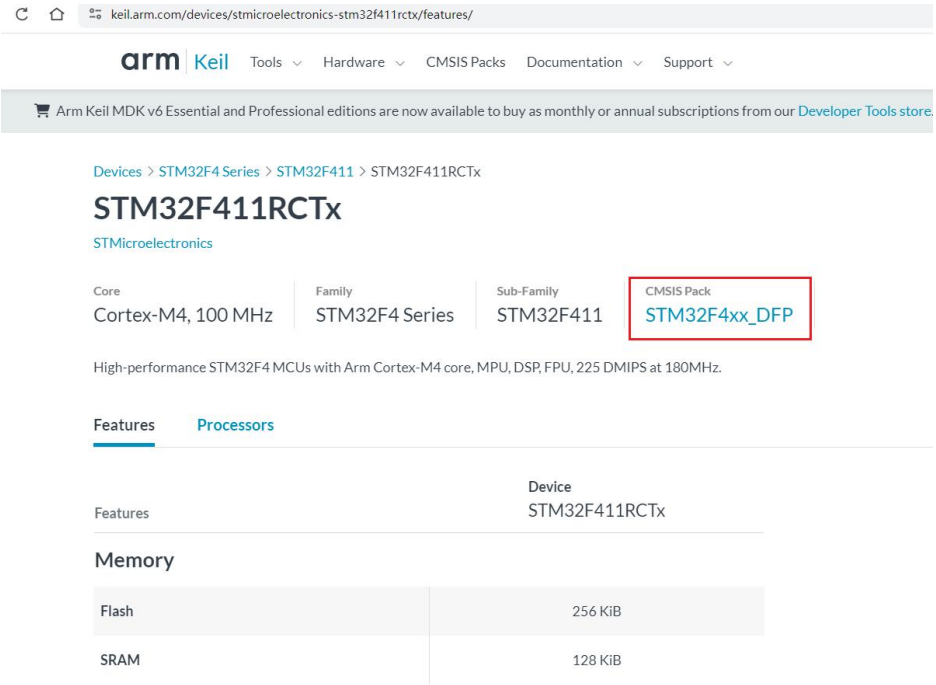

安装结束。

### 3.1.2 器件支持包安装

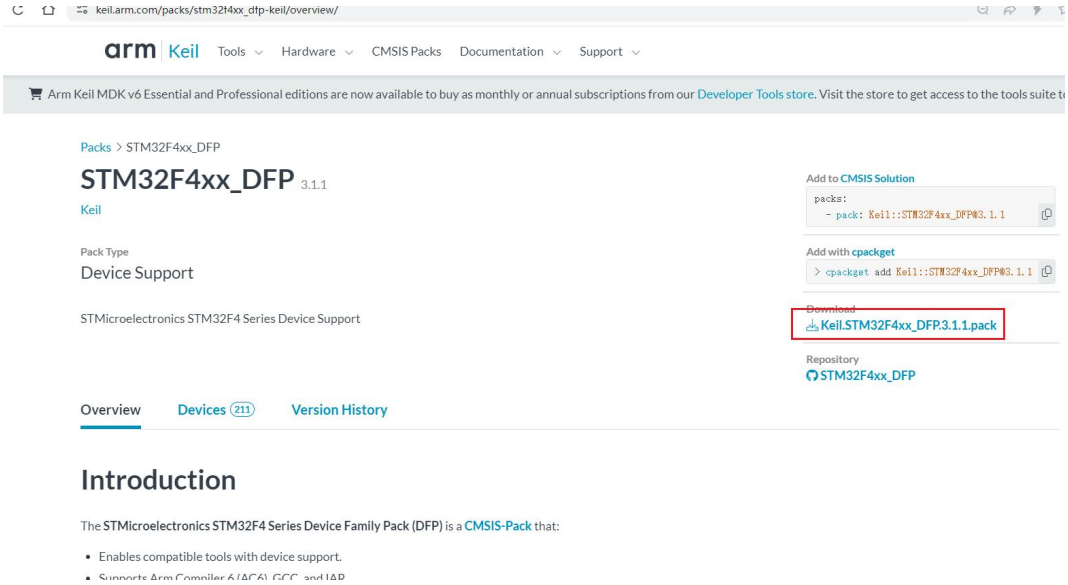
器件支持包下载：https://www.keil.com/dd2/pack/



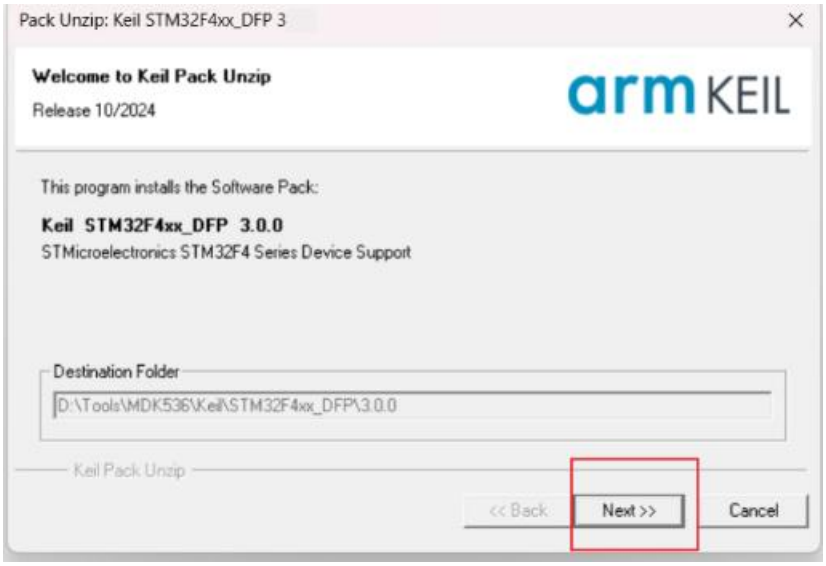

搜索需要用到的芯片的器件支持包



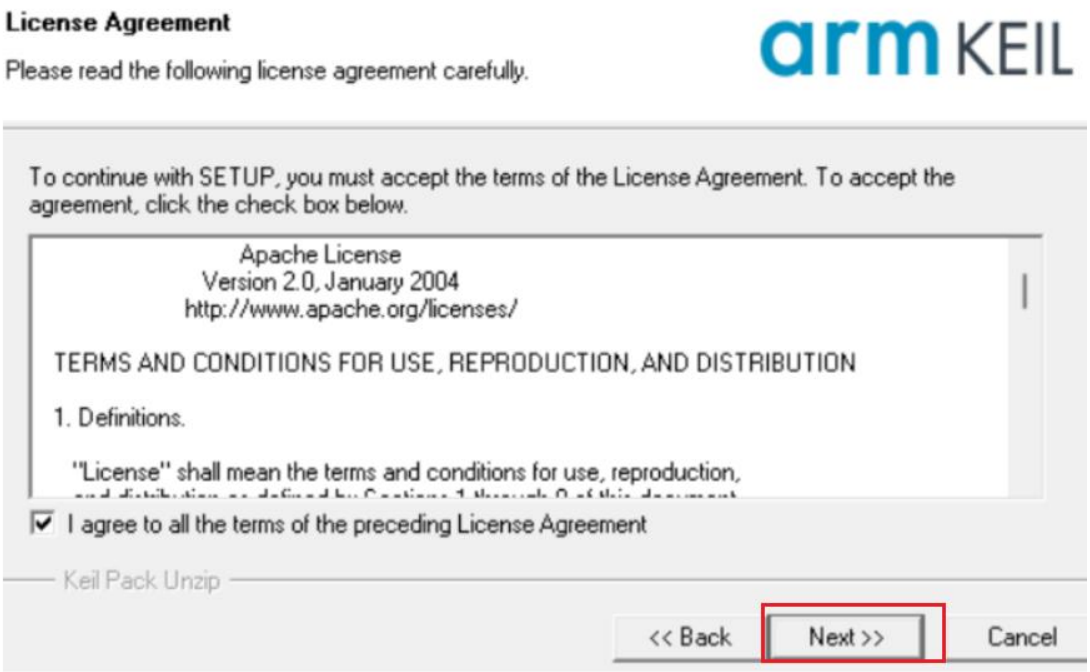

点击后进入



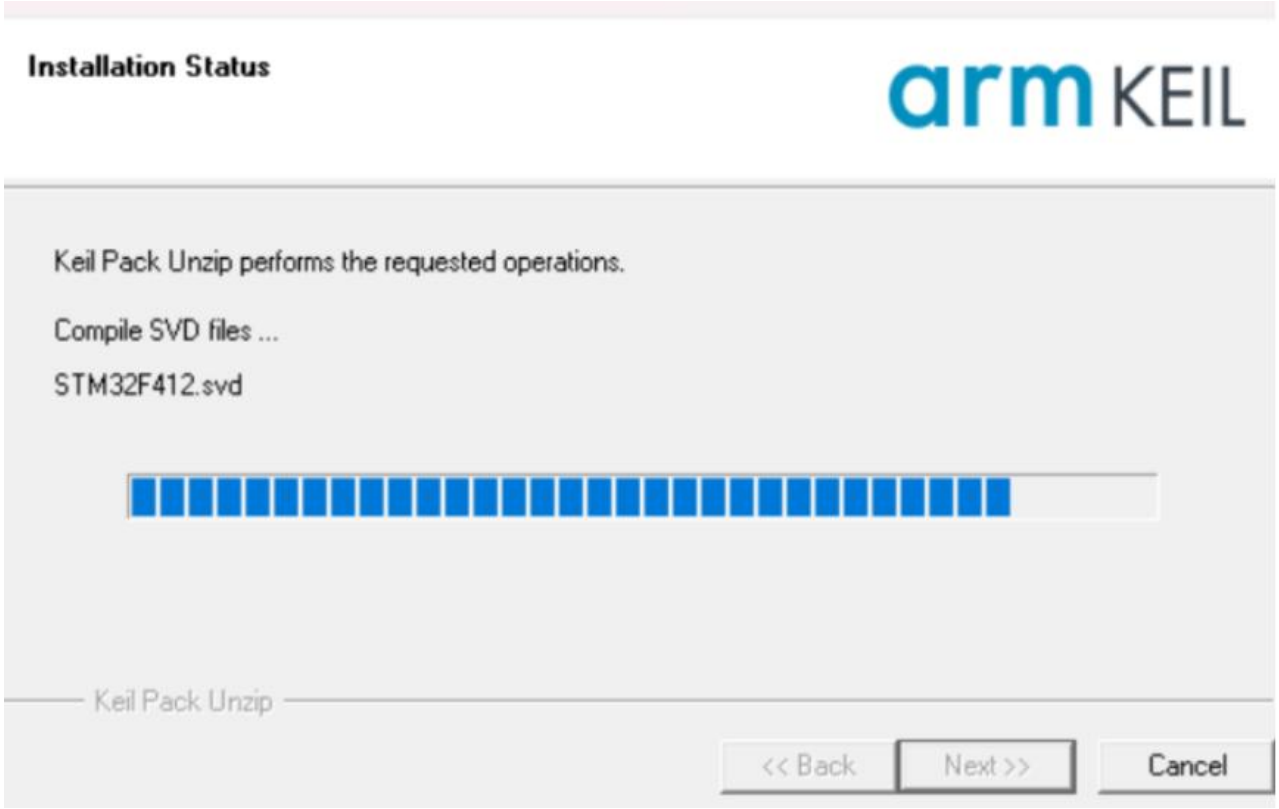

点击进行下载支持包，下载完成后双击下载的 pack




点击 next 进行安装




勾选 I agree，点击 next 继续进行安装



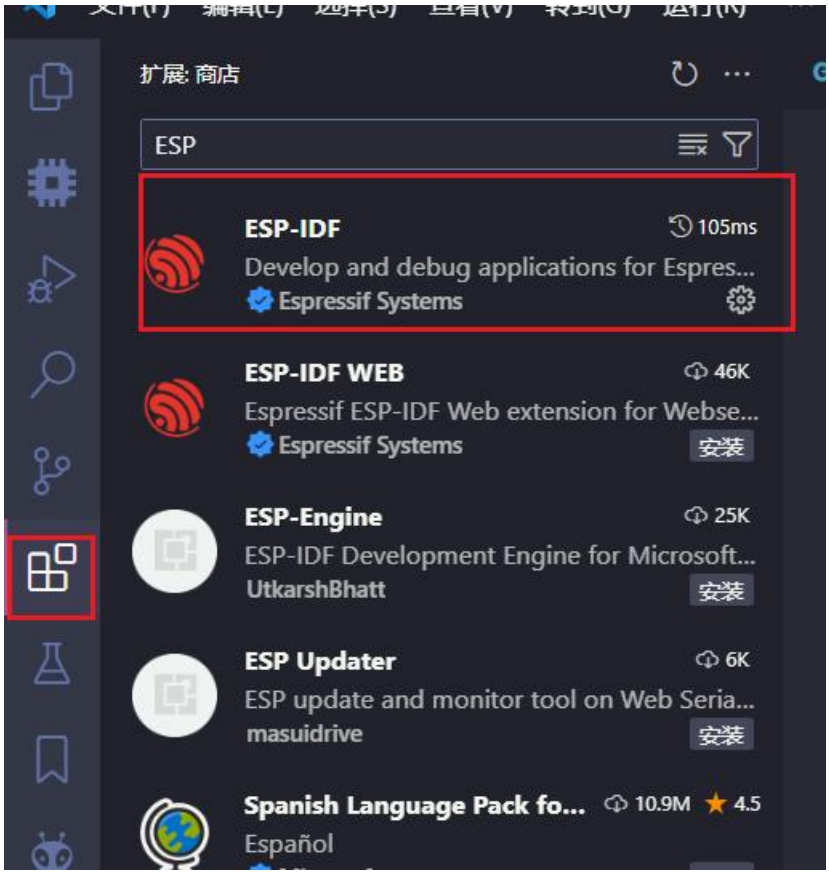

等待安装完成后，下一个界面点击 finsh 即完成安装

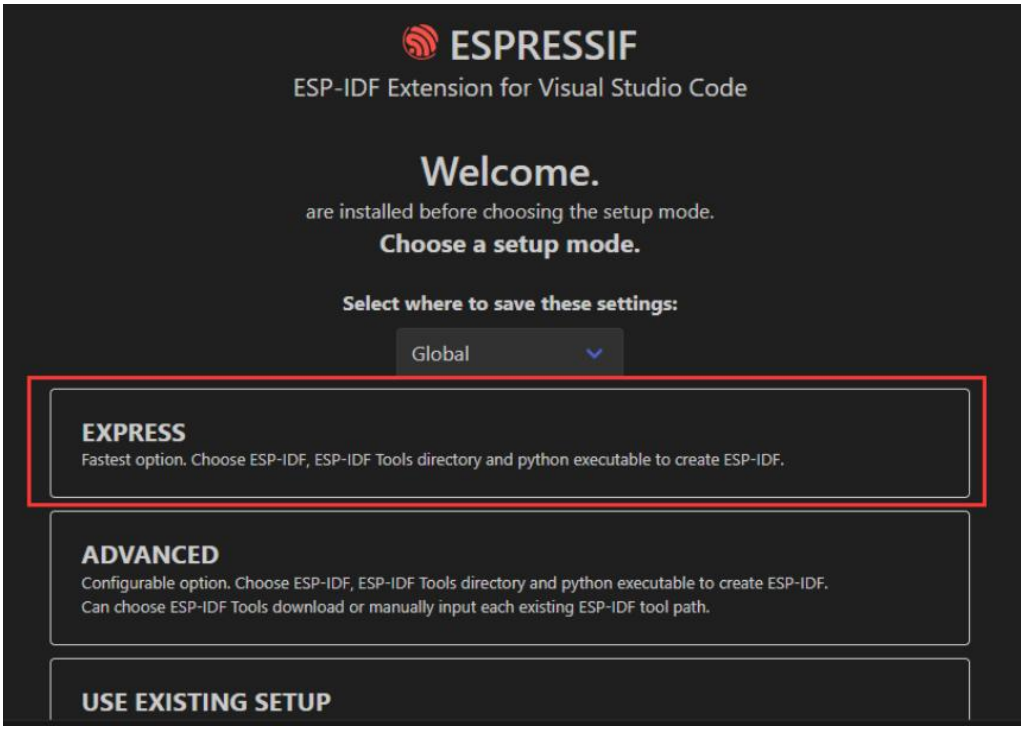
## 3.2 ESP32 环境安装

Vscode 软件安装包地址：https://code.visualstudio.com/

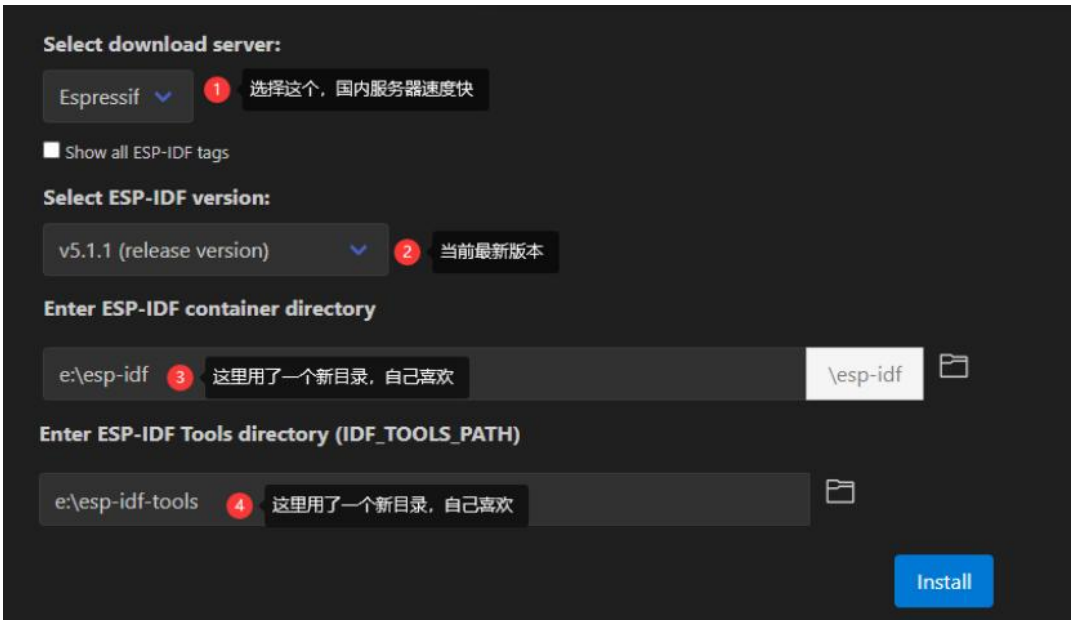
选择对应的版本下载安装即可，这里具体讲解相关编译和烧录插件的安装，以下内容进行的前提是基于 vscode。

### 3.2.1 ESP-IDF 插件安装

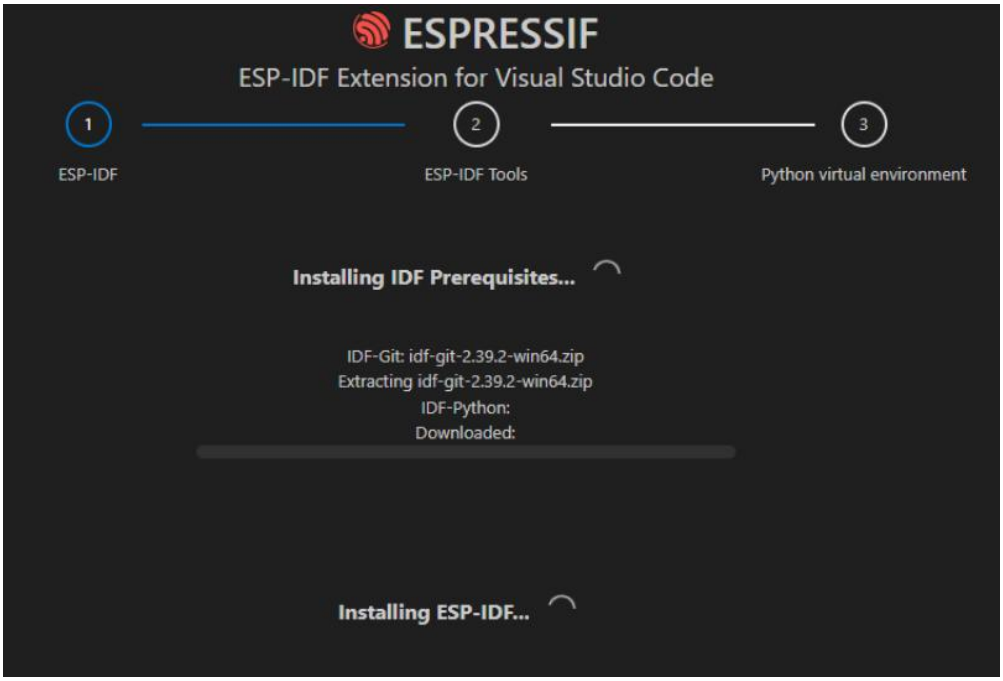


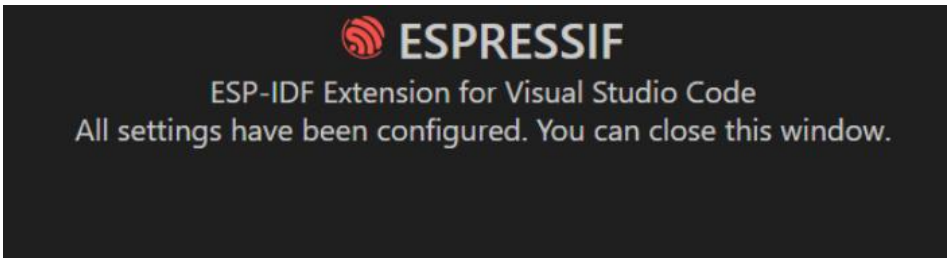

按下 CTRL+shift+P 打开设置搜索，输入 esp-idf configure esp-idf extension，点击 EXPRESS




根据需要选择目录，之后点击 Install



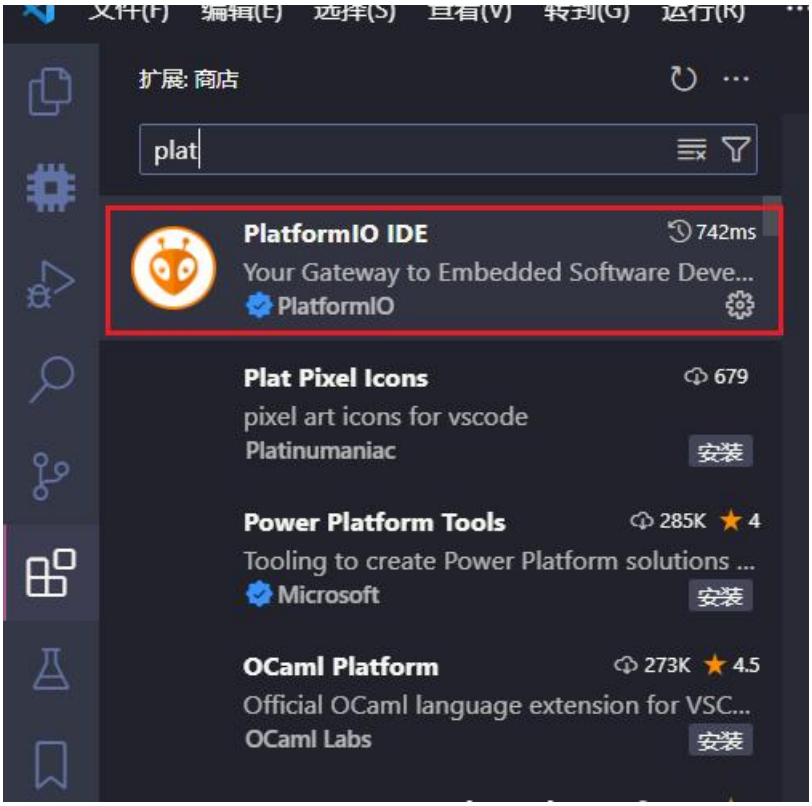

等待安装完成



### 3.2.1 platformIO 插件安装

PlatformIO（也称 PIO）是一款开源且跨平台的嵌入式系统与产品程序的开发工具，可用于多种嵌入式架构与平台的开发，支持多种嵌入式平台，例如 STM32、STM8、ESP32、Amtel AVR、ESP8266、8051 等。

在 vscode 中搜索该插件，下载即可。




## 3.3 ASPRO 环境安装

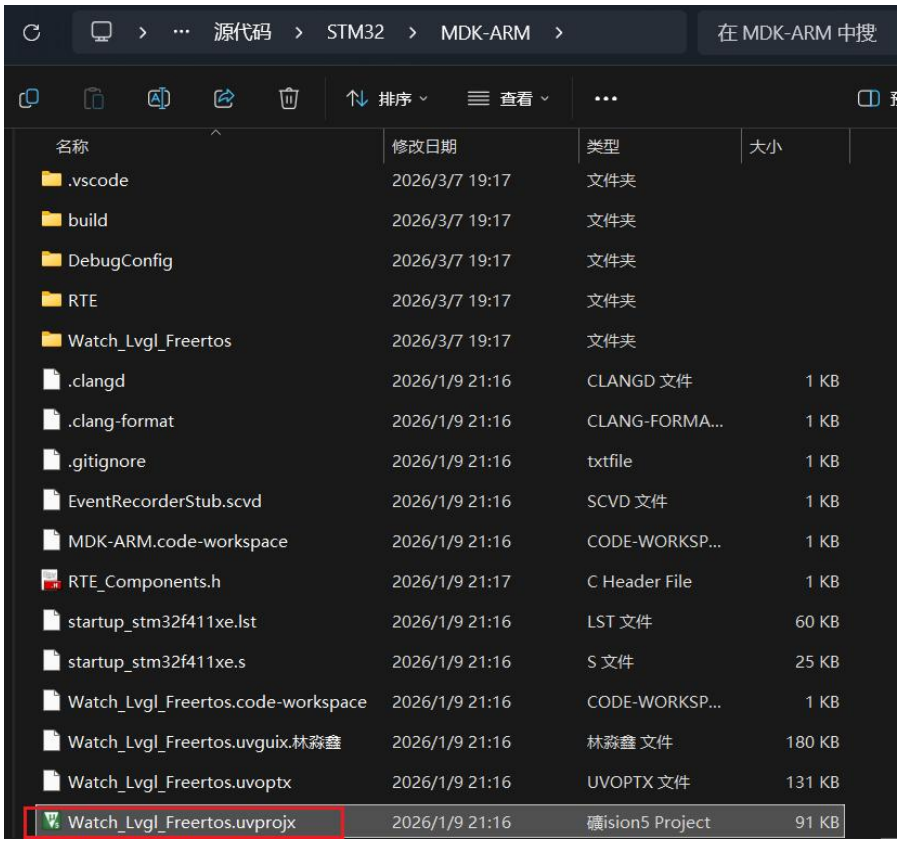
具体的软件安装链接：http://www.twen51.com/new/twen51/index.php

不安全 | twen51.com/new/twen51/index.php
首页
在线编程
发现
课程
学校
资料
软件
社区
登录
注册
Block
天问Block 单片机开发利器，支持STC全系列8位单片机，32位ARM和RISC-V内核。
天问Block 下载

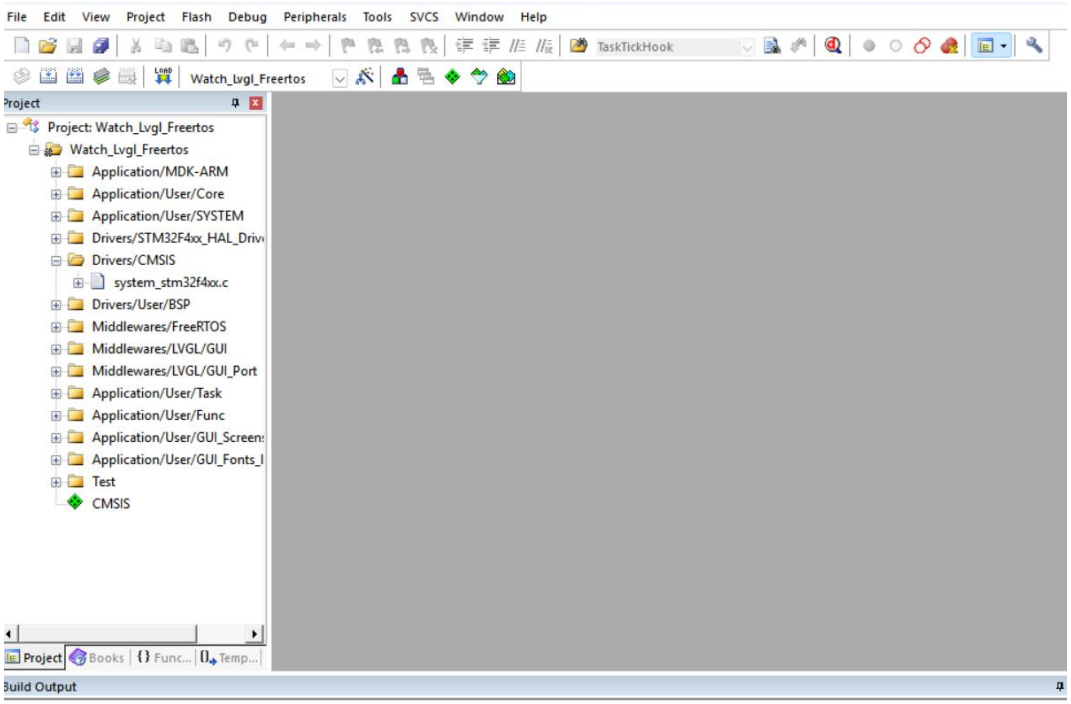
# 4 烧录部署

## 4.1 STM32 烧录

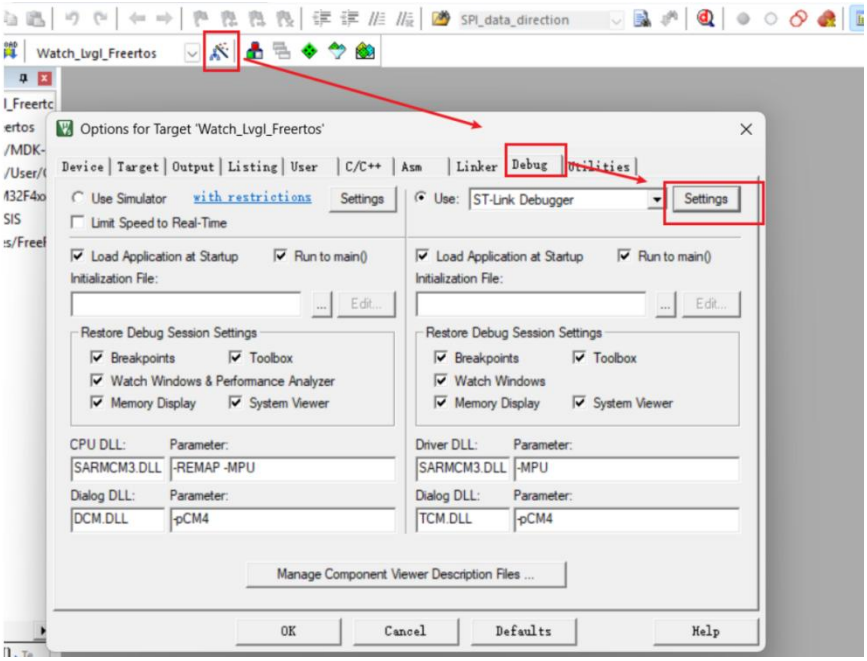
连接 ST-Link 与 STM32 开发板，在 STM32 的代码中找到该文件：

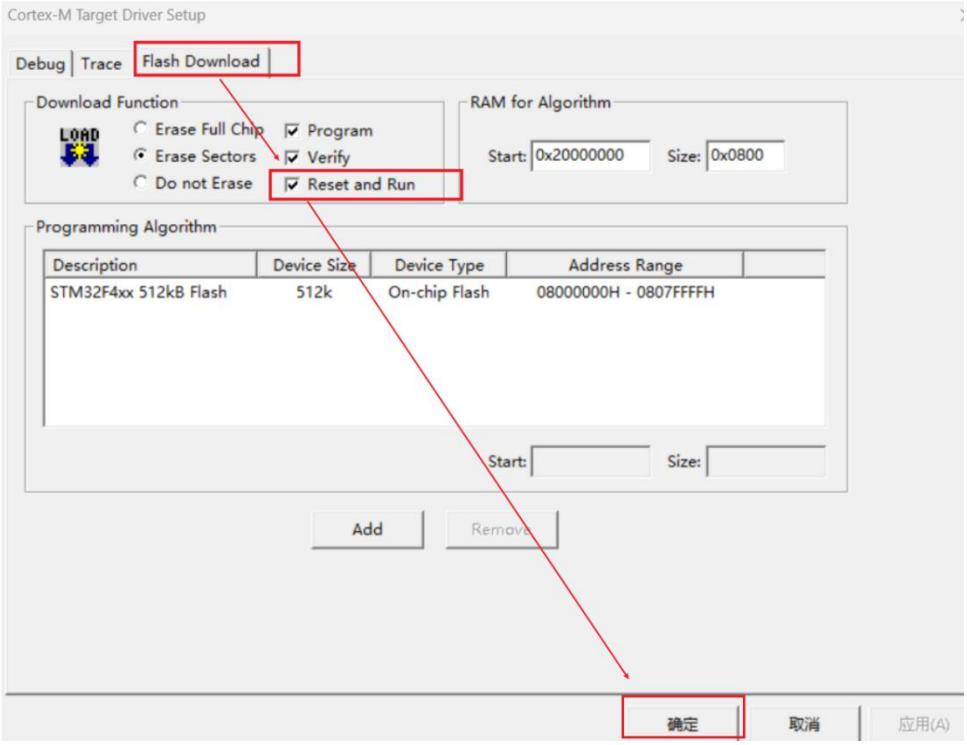



双击打开 keil 工程

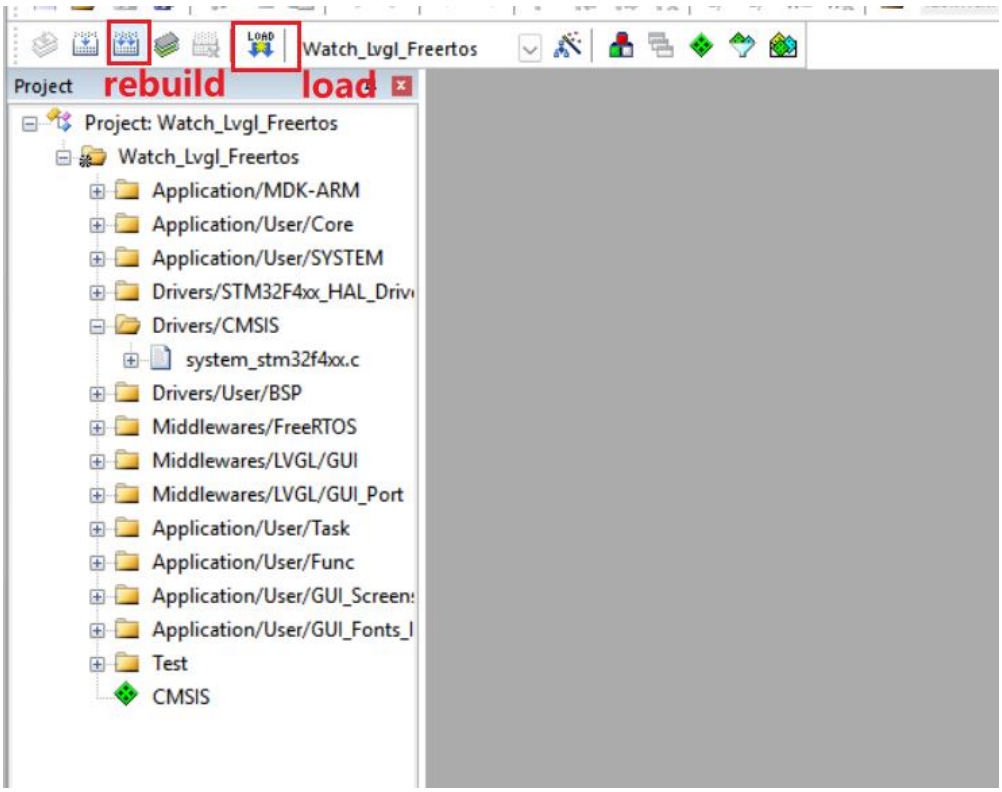

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
TaskTickHook
Watch_Lvgl_Freertos
Project
Project: Watch_Lvgl_Freertos
Watch_Lvgl_Freertos
Application/MDK-ARM
Application/User/Core
Application/User/SYSTEM
Drivers/STM32F4xx_HAL_Driv
Drivers/CMSIS
system_stm32f4xx.c
Drivers/User/BSP
Middlewares/FreeRTOS
Middlewares/LVGL/GUI
Middlewares/LVGL/GUI_Port
Application/User/Task
Application/User/Func
Application/User/GUI_Screens
Application/User/GUI_Fonts_I
Test
CMSIS
Project Books {} Func... Temp...
Build Output

打开后的界面如上


SPI_data_direction
Watch_Lvgl_Freertos
Options for Target 'Watch_Lvgl_Freertos'
Device Target Output Listing User C/C++ Asm Linker Debug Utilities
Use Simulator with restrictions Settings
Limit Speed to Real-Time
Use: ST-Link Debugger Settings
Load Application at Startup Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints Toolbox
Watch Windows & Performance Analyzer
Memory Display System Viewer
Watch Windows
CPU DLL: Parameter:
SARMCM3.DLL -REMAP -MPU
Driver DLL: Parameter:
SARMCM3.DLL -MPU
Dialog DLL: Parameter:
DCM.DLL -pCM4
TCM.DLL -pCM4
Manage Component Viewer Description Files ...
OK Cancel Defaults Help




查看复位和启动是否勾选

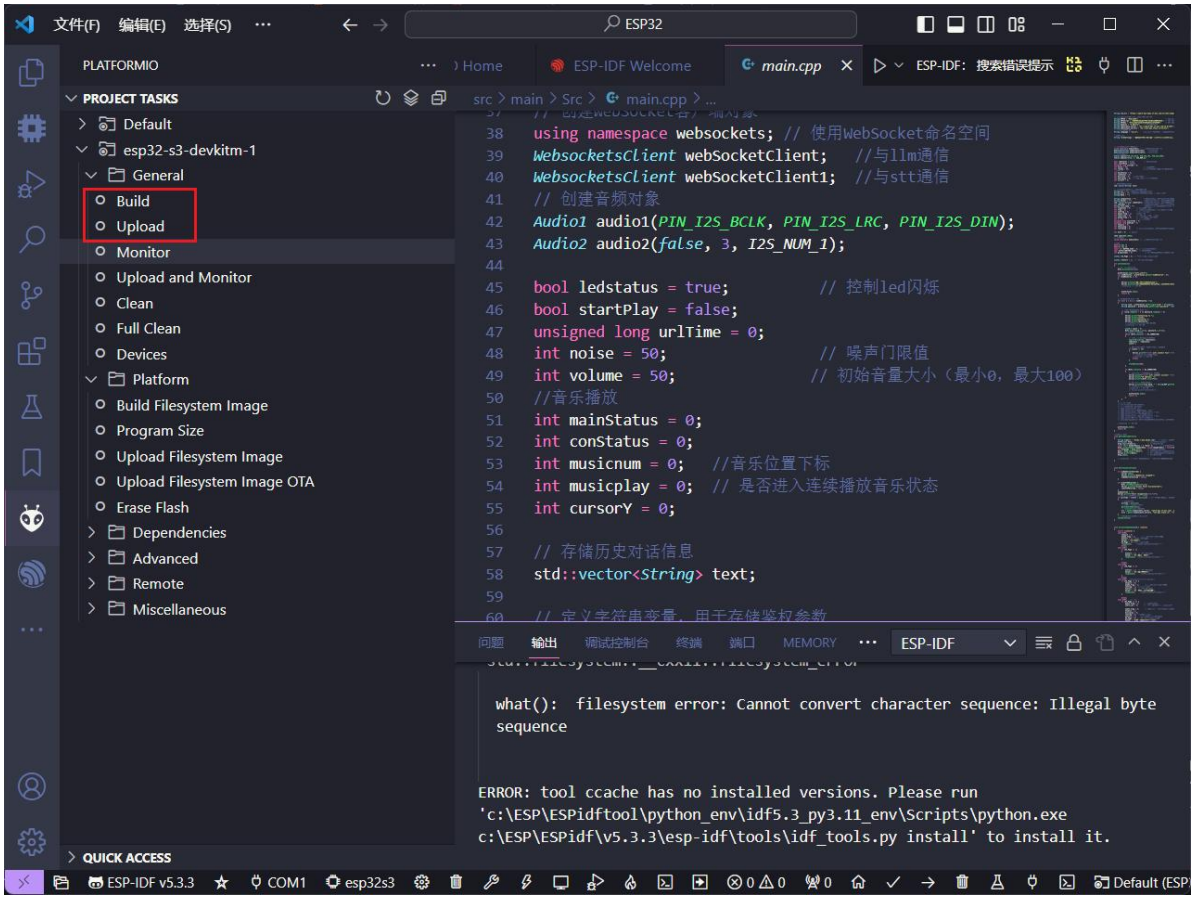



确保 STM32 开发板通过 ST-Link 与电脑连接后，先点击 rebuild 进行源程序编译，之后点击 load 进行烧录

## 4.2 ESP32 烧录

打开下载好的 vscode 软件

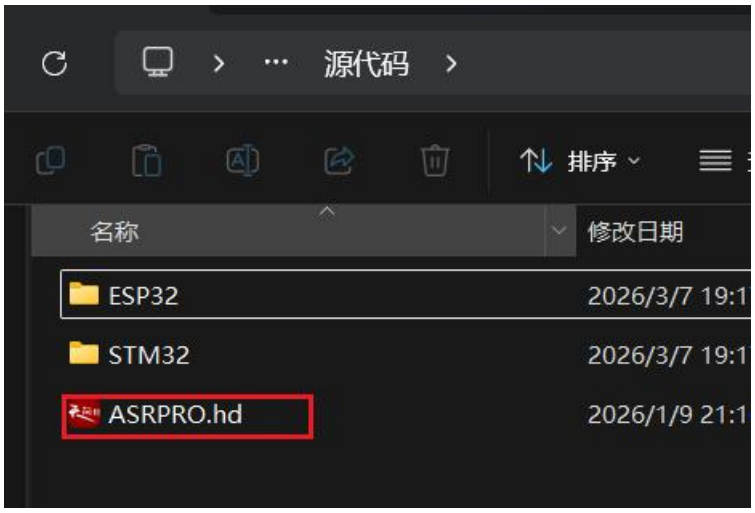
打开 ESP32 的源程序文件夹，打开后 platformIO 插件会自动识别




识别后的界面如上，build 是编译程序，upload 是将编译后的程序烧录进 ESP32

## 4.3 ASRPRO 烧录

下载好官方提供的天问 Block 后



双击该文件

确保 ASRPRO 和电脑相连接后，检查连接的 COM 口是否对应，之后点击 2M 编译下载即可。